Privacy – Enhanced Synthetic Data Generation for Autonomous Vehicle Training

R26-CS-010

Individual Thesis

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Individual Component: Utility Evaluation

Group ID: R26-CS-010

BSc (Hons) Degree in Information Technology Specialized in Cyber Security

Department of Computer Systems Engineering

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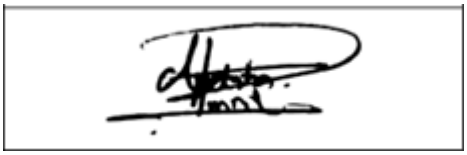
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Supervisor certification. The above candidate is carrying out research for the undergraduate Dissertation under my supervision.

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Mr. Dinith Primal	03/10/2026	

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I also extend my appreciation to my group members responsible for the privacy calibration, dataset generation, and risk and compliance components of this project for their commitment, cooperation and collaboration in developing the broader research pipeline of which this component forms a part. Finally, I acknowledge the open-source community (Ultralytics, PyTorch, CARLA, torchmetrics) for providing the tools that made this evaluation possible.

Abstract

Perception systems for autonomous vehicles rely on high amounts of labeled driving imagery that historically is sourced from actual footage with identifiable content like pedestrian faces and car license plates. There is an alternative approach in generating synthetic data for privacy preserving data training. However, privacy preserving is insufficient in proving the applicability of generated data. This report describes the utility evaluation module for a four staged privacy preserving synthetic data pipeline, where a differentially private (Laplace mechanism) calibration stage prepares CARLA simulator for generation of synthetic driving footage. A security validated evaluation workflow is discussed, which includes decryption, integrity verification, statistical data poisoning check, exploratory data analysis, and YOLOV8 object detection model training and evaluation on a five towns, 10,000 frames synthetic dataset. Overall and per class mean Average Precision (mAP) values are provided for a synthetic test split of the dataset (mAP50-95 = 0.2273). Performance comparison is provided of synthetic and real data trained models on real data images (KITTI benchmark), showing the sim to real gap with the former being able to achieve only 10.46% performance of the latter (mAP50=0.0889 vs. 0.8500). A Fréchet Inception Distance (FID) score is provided for a quantitative measure of statistical similarity between synthetic and real image distribution (FID = 115.99). The results show that the privacy preserving synthetic dataset is capable of training a functional AV perception model, while quantifying the performance disparity across object classes as a function of dataset composition rather than model failure.

Keywords: Privacy preserving synthetic data, CARLA simulator, YOLOv8, Mean Average Precision, Fréchet Inception Distance

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List of Abbreviations

Abbreviation	Full Form
AES	Advanced Encryption Standard
AP	Average Precision
AUC	Area Under the Curve
AV	Autonomous Vehicle
CARLA	Car Learning to Act (open source driving simulator)
DP	Differential Privacy
DP-SGD	Differentially Private Stochastic Gradient Descent
FID	Fréchet Inception Distance
GDPR	General Data Protection Regulation
GPU	Graphics Processing Unit
IoU	Intersection over Union
KITTI	Karlsruhe Institute of Technology and Toyota Technological Institute (dataset)
mAP	mean Average Precision
PDPA	Personal Data Protection Act (Sri Lanka)
SHA-256	Secure Hash Algorithm 256 bit
YOLO	You Only Look Once (object detection architecture)

Table 1: List of Abbreviations Used in this Report

1. Introduction

1.1 Overall Project Context

The Autonomous Vehicle (AV) perception system uses a vast amount of data in the form of images that have been manually annotated in order to detect the various users on the roads. Normally, this kind of dataset is gathered from the videos of actual driving scenarios and thus such dataset will contain the personal data of individuals such as facial information of pedestrians and license plates of vehicles. This use of personal data for the training of AI models poses serious legal issues in terms of data protection law like the General Data Protection Regulation (**GDPR)** and Sri Lanka's Personal Data Protection Act (**PDPA)**.

Simulation of driving data has been put forward as a possible solution, which would replace the collection of data by creating synthetic data through simulation environments. Nonetheless, this process does not necessarily mitigate the issue of privacy completely since the simulation environment will have to be calibrated using statistics obtained from actual driving practices (such as traffic density and speed). Thus, the problem of privacy is simply transferred to another phase in the data processing chain.

This is tackled through the group study by use of four stages privacy preserving synthetic data pipeline, as known in table below.

Stage	Owner	What it does
1. Privacy Calibration	IT22309556	Adds mathematical noise (the Laplace mechanism) to real world statistics before they reach the simulator, and applies DP-SGD during downstream model training.
2. Dataset Generation	IT22541284	Configures the CARLA simulator using the privatized statistics and generates synthetic driving footage across five towns.
3. Utility Evaluation	IT22110220 (this report)	Measures whether the resulting synthetic dataset is actually good enough to train a real AV perception model.
4. Risk and Compliance	IT22066916	Attacks the pipeline and maps the implemented controls to relevant GDPR articles.

1.2 Individual Component Context

Utility Evaluation, as detailed in this report, is the third step in this pipeline, which aims to evaluate whether the generated synthetic data has enough utility to be used for training an autonomous vehicle perception model.

1.3 Problem Statement

Privacy is straightforward to achieve if usefulness is disregarded entirely a dataset composed of pure noise would satisfy strong privacy guarantees while being completely unusable for model training. The Utility Evaluation component addresses a central limitation in privacy preserving data generation research. Privacy guarantees alone do not establish practical value. Without an empirical utility measurement, the group's broader project could claim privacy protection but could not claim that the resulting synthetic data is usable leaving the "utility" half of the privacy utility trade off, which is central to the project, unaddressed.

A further complication is that the privacy preserving synthetic dataset available for evaluation was generated at a single calibration privacy parameter (epsilon = 1.8), rather than a cross multiple privacy budgets, since the pipeline's architecture products privatized configuration parameters upstream of scene generation rather than a directly modifiable dataset. This report addresses how a complete and defensible utility evaluation can still be conducted under this constraint.

1.4 Research Gap

The current state of research on privacy preserving datasets for AVs concerns two distinct issues: how to ensure privacy (for instance, by using differential privacy at the input level before the simulator [1]), and how realistic the images are (for example FID[5]). It is clear from section 2.5 that neither privacy assurances nor image realism guarantee that a particular dataset will train a functioning perception model; an empty set of random noises satisfies the former but not the latter.

As seen from the literature discussed in Chapter 2, there are four remaining gaps for building a pipeline, where a simulator is calibrated by DP statistics:

- **Missing utility evidence.** There is no measurement of how an object detector trained only using DP calibrated CARLA data is effective, especially when it is applied to images from real world imagery.
- **Aggregate metrics hide class level failure.** Overall mAP metric might mask really poor performance of the underrepresented and small object detection classes like pedestrians and cyclist [7], the most critical in terms of safety of all.
- **Unknown relationship between realism and utility.** It is not known whether FID scores predict the effectiveness of detection in the case of this type of dataset.
- **Unaudited external data.** Labeling done through another component in the pipeline is considered valid implicitly yet label level data poisoning is possible without changing an image [9].

This individual metric makes up for this deficiency by performing a mAP comparison through four experiments across different domains in relation to an independent, real world benchmark (KITTI) an FID metric, per class and per size reporting and an integrity check on the labels, whose detector is tested empirically. All of these together form the utility side of the privacy utility tradeoff for the group's process.

1.5 Research Aim

In order to conduct an empirical test of the effectiveness of training a functionally capable autonomous driving object detection system using privacy preserving synthetic driving data calibrated with differentially private parameters, as compared to training it with real world driving data.

1.6 Objectives

1.6.1 Main Objective

The primary goal of this study is to measure, through rigorous experimentation, whether a privacy preserving synthetic dataset generated by the group's CARLA based pipeline is capable of training an object detection model (car, pedestrian and cyclist) to a standard comparable with a model trained on real world data and to quantify the resulting utility cost in defensible, reproducible terms.

1.6.2 Specific Objectives

1. To securely decrypt and verify the integrity of the received synthetic dataset using AES 256 decryption and SHA 256 manifest verification.
2. To statistically explore the dataset's class distribution and object size composition prior to training, in order to anticipate and explain downstream performance patterns.
3. To implement and empirically validate a data poisoning detection mechanism that audits label integrity before training, addressing the security component required of this project.
4. To convert the dataset into YOLO training format and train a YOLOv8n object detection model on the synthetic data, reporting overall and per class mean Average Precision (mAP) on a held out test split.
5. To establish an independent real world baseline (the KITTI benchmark) and conduct a four experiment cross-domain evaluation to quantify the sim to real utility gap.
6. To compute the Fréchet Inception Distance (FID) between the synthetic and real image distributions as a complementary, task-independent measure of dataset realism.

1.7 Research questions and hypotheses

These research questions guide the work:

i. **RQ1.** Is it possible for an object detection model (YOLOv8), which is trained solely on the privacy preserving synthetic dataset, to detect cars, pedestrians and cyclist in the held out synthetic images?

ii. **RQ2.** What portion of the detection performance is maintained by the model that is trained on synthetic images compared to the one that is trained on real images in the real world imagery (KITTI)?

iii. **RQ3.** How is the maintenance of the utility divided among various object categories and object sizes?

iv. **RQ4.** Is there any bidirectional transfer learning domain gap between CARLA rendered and real imagery?

v. **RQ5.** Is the task level detection utility predicted by the image level realism score (FID)?

vi. **RQ6.** Is the generated dataset free from label manipulation?

Three research hypotheses go along with these queries. **H1:** the amount of data is sufficient for training an adequate detector, meaning that the performance on a synthetic test set is definitely better than random guessing. **H2:** the performance on actual images is worse for a detector trained on synthesized data compared to a detector trained on actual data. **H3:** this difference is not equal for all classes but depends on their frequency and size.

1.8 Scope limitations and assumptions

Scope: - This report addresses stage 3 (Utility Evaluation) of the four stage pipeline undertaken by the group. It involves the use of the 10,000 frame synthetic CARLA dataset that contains five towns and is of 800x600 dimensions with RGB frames and YOLO labels, three classes of objects (cars, pedestrians and cyclists), YOLOv8n detection model, the KITTI 2D Object Detection benchmark as real world baseline, mAP50 and mAP50-95 as the detection metrics, FID as the realism metric and a label integrity audit. Stage 1 (Privacy calibration and DP – SGD) and Stage 2 (Dataset Generation) and Stage 4 (Risk and Compliance) were done by other members of the group and are beyond the scope of this thesis.

Limitation: - The data set was created at a sole privacy value of epsilon = 1.8 and therefore cannot yield any privacy versus utility graph for different privacy budgets. CARLA (800x600, 4:3) and KITTI (approximately 1242x375) images have differing resolutions and aspect ratios, hence causing some portion of the sim to real transfer problem. Training and testing was done on a CPU (Intel Core i5 – 8250U), thereby restricting the study to the small version of YOLOv8 and using a training budget of 50 epochs. Every reported result has been generated from a sole trained model of each domain with a fixed seed (42).

Assumption: - (i) KITTI is an adequate proxy for realistic driving images and as such does not suffer from circularity due to the lack of correlation with the statistics provided by Waymo. **(ii)** An unencrypted file with a matching SHA – 256 hash to the manifest file is an exact copy of the one generated during the generation phase. **(iii)** The provided labels are ground truth unless the poisoning audit proves otherwise. **(iv)**The mapping of KITTI categories to the three classes in this project is semantically equal to the synthetic classification schema. **(v)** The guarantee of differential privacy is stage 1 is assumed.

1.9 Individual contribution

The personal contribution of the author to the group project is the designing, implementation and execution of the whole utility evaluation phase that includes:

- Decryption component (AES - 256 – CBC) with SHA 256 manifest validation of the downloaded dataset (40,000 images, five towns)
- Exploratory data analysis for class balance, empty images and objects sizes and explanation of per class downstream performance
- Data poisoning detection algorithm (structural and statistical validation), as well as proof of concept injection
- Conversion scripts for preparation of the synthetic dataset and KITTI baseline in the same YOLO format and splitting into 70/15/15 with the same seed
- Training of two YOLOv8n models (synthetic and KITTI datasets) with the same hyper parameters, four experiments across domain evaluation module
- FID calculation module and analysis of the mismatch between FID and mAP
- Conclusions that provide data level privacy cost for the privacy utility trade off argument of the group

The modules possessed by other group members (Privacy Calibration, Dataset Generation, Risk and Compliance) are mentioned just enough for understanding the connection between this module and the others.

1.10 Thesis structure

This thesis will be structured in the following manner.

Chapter 2 (Literature Review and Related Work) the topic that will be covered include privacy risk of AV training dataset, differential privacy and Laplace mechanism, data poisoning, CARLA synthetic data, YOLO and mAP evaluation and FID and ends off with gap synthesis.

Chapter 3 (Research Methodology) will discuss about the methodology used, the datasets used, the trainings conducted, cross domain and FID methodology, the metrics used, exploratory analysis, security and ethical issues and the reproducibility.

Chapter 4 (Product and Solution Design) the requirements, the role of the component in the whole system and the design of the solution will be discussed.

Chapter 5 (Implementation, Integration and Testing) discusses the development environment, implementation issues, integration with other parts, testing results and solution operation.

Chapter 6 (Results and validation) discussed the detection results, FID results, and poisoning audit results as well as integration results and conclusions drawn from them.

Chapter 7 (Discussion) discusses the results and their interpretation, comparison with literature, research and product contributions, limitations and implications.

Chapter 8 (Conclusion and Future Directions) summarizes the thesis, explains how each objective is accomplished, summarizes individual contributions and suggests future works. The report ends with references, glossary and two appendices.

2. Literature Review and Related Work

2.1 Review method and scope

An increased in the usage of big annotated datasets for autonomous vehicle studies has created an urgent need to pay attention to the privacy risks associated with using real – world driving videos. This chapter presents the review of the literature devoted to the issue of privacy regulation of driving data, differential privacy and creation of synthetic data for self – driving cars, object detection and data poisoning, after which the research gap will be identified.

2.2 Domain and theoretical foundations

2.2.1 Privacy risk in Autonomous Vehicle Training Data

The actual driving data will naturally contain personally identifying information, in the form of faces of pedestrians and license plates of vehicles. The acquisition and use of data for the training of AI models is regulated by a number of legislative acts such as GDPR (General Data Protection Regulation) in Europe [11] and the Personal Data Protection Act in Sri Lanka [12]. This creates substantial liability risks for organizations involved in gathering and processing of the driving data, thus driving the need for privacy preserving methods.

2.2.2 Differential Privacy and the Laplace Mechanism

The concept of differential privacy offers a mathematical way of ensuring that when statistical information about the data set is released, inference to any individual data record in that data set should be minimally affected by that release [1]. One of the fundamental approaches in differential privacy is the use of the Laplace mechanism, which works through the process of adding appropriate statistical noise before the release of the aggregate values. This is done within the larger group pipeline, whereby real world traffic statistics are first modified before setting up the simulation environment.

2.3 Related research

2.3.1 Data Poisoning in Machine Learning Pipelines

The learning pipelines using the data that is generated by some other external or independent third party are vulnerable to attack through poisoning the data by adding poisoned labels in such a manner that the learning model gets corrupted in its functioning without making any change to the real images [9]. Bounding boxes that have been altered will teach the model to locate objects in the wrong places, while altered annotation labels will teach the model to identify objects that do not even exist. The above discussed threat is very much relevant to the multi component pipeline approach used in our case study.

2.4 Existing systems products and techniques

2.4.1 Synthetic Data Generation for Autonomous Driving (CARLA)

The CARLA simulator is an Open source platform created specifically for autonomous driving research, allowing traffic, weather and sensor configuration [2]. The use of synthetic data, which can be created, using simulators like CARLA, has been suggested as a privacy preserving method compared to data collection from the real world, where there are no real people involved. But, as real world statistics must be used in order to achieve realistic simulations, the issue of privacy is simply postponed to the calibration process of the simulator – as discussed in this report.

2.4.2 Object Detection Evaluation with YOLO and mAP

One of the most commonly applied real time object detection models include You Only Look Once (YOLO), which are designed to solve the detection task as one regression task across the entire image [3]. Detection accuracy of model performance is usually measured by mean Average Precision (mAP) values, which is a standard measure of performance for object detection in widely adopted benchmarking tasks like the Microsoft COCO dataset [6]. Class by class evaluation is advised for use in case of class imbalance since aggregation metrics like mAP could hide significantly lower performance of detection on less represented classes [7].

2.4.3 Evaluating Synthetic Data Realism with FID

Fréchet Inception Distance (FID) is one of the metrics that has been popularly used to measure statistical similarity between two collections using feature representations from a pre trained Inception V3 model [5]. The smaller the FID value, the more similar the distributions of the image collections under comparison. FID has been shown in the literature of generative models as a task independent metric to the task specific metrics, providing justification for including it alongside mAP.

2.5 Gap synthesis and conceptual framework

Based on the literature reviewed above, it can be concluded that **(i)** differential privacy algorithms may be used upstream from the production of synthetic datasets. However, it means that the risk of privacy has been transferred rather than removed. **(ii)** The performance of object detection should be measured for each class, rather than just overall, because there is an issue of class imbalance; and **(iii)** FID is a meaningful metric to measure realism regardless of the particular task at hand. However, it does not provide evidence regarding whether a specific privacy – preserving synthetic driving dataset will have sufficient utility for training a practical perception model compared to real world data. Neither does it use a combination of mAP and FID metrics with an integrity check of the data.

3. Research Methodology

3.1 Research Design

This chapter outlines the methods adopted for designing, developing and evaluating the utility of the privacy preserving synthetic dataset. The method adopted is focused on the entire pipeline of conducting a utility evaluation from data manipulation to training the models across the domains using a systematic procedure of comparing the synthetic and real world datasets.

3.1.1 Research Approach

The research was conducted using an empirical experimental method. The evaluation was carried out by comparison of detection efficiency in four combinations of train / test data sets defined systematically, as well as by a distribution similarity measure (FID) and a security validation procedure (data poisoning detection). The steps of the methodology included:

- Environment setup and checking
- Securely decrypting and verifying the integrity of the received dataset
- Exploratory data analysis (class distribution, object size distribution)
- Detecting data poisoning, including testing the detector empirically
- Converting the dataset to YOLO training format
- Training and evaluating a YOLOv8n model on the synthetic dataset
- Collecting and converting an independent real world dataset (KITTI)
- Cross domain evaluation across four train/test combinations
- Computation of the Fréchet Inception Distance (FID)
- Result aggregation and analysis

3.2 Dataset Description

The synthetic dataset used in this evaluation was generated by the dataset generation component of the group's pipeline using the CARLA simulator, configured with privatized calibration parameters. The dataset spans five distinct CARLA town environments, comprising a total of 10,000 annotated frame sets. For each frame, four sensor modalities are captured: an RGB camera image, a corresponding depth map, a semantic segmentation image and a single channel class mask. All images are captured at a native resolution of 800 x 600 pixels.

Ground truth object annotations are provided in YOLO format, with each label file containing one line per detected object, structured as a class identifier followed by normalized centre – x, centre – y, with and height coordinates (all values scaled to the range [0,1]). Three object classes are annotated: car (encompassing vans, trucks and buses), pedestrians and cyclist (encompassing motorcycles). Frames containing no detectable objects are represented by empty label files; these

are treated as valid data points rather than errors and all data processing scripts developed for this component were designed to handle them without failure.

To preserve confidentiality during transfer, all sensor images (with the exception of label files) were provided in AES 256 encrypted form, accompanied by a SHA 256 integrity manifest for each town, used to verify successful and loss less decryption prior to any downstream processing.

3.3 Environment Setup

A dedicated and isolated Python environment (Python 3.10 / 3.11) was constructed with the use of conda for the purposes of reproducibility. The following software packages have been installed: Ultralytics (with YOLOv8), PyCryptodome (for AES decoding), Pillow, Numpy, Pandas and Scipy (used for data processing and exploration), as well as torchmetrics with its extra modules for images (which provide an FID implementation). PyTorch has been installed with CUDA support where available. CUDA capability has been checked before starting the training process, which took place on CPU where there was no GPU driver.

3.4 Secure Decryption and Integrity Verification

All encrypted files consisted of an initialisation vector of 16 bytes and were then encrypted using AES 256 CBC with PKCS7 padding. A decryption pipeline was developed for the decryption of the original image data contained in all files in each of the five simulation environments. To prevent accidental disclosure of the key during deployment, the key was loaded at runtime from a managed configuration file instead of being hard coded in the source code, and the managed configuration file itself was not under version control.

Post decryption, the SHA 256 hash of each file was generated and compared against the reference manifest that came with the dataset, enabling detection of any corrupted file while preventing its use in later processing. The unencrypted annotation files (bounding boxes) were simply copied without change, while the original encrypted files were retained to enable repeatable decryption of all files without having to request the dataset again. This approach yielded 100% success of decryption, where all hashes matches those from the manifest.

3.5 Exploratory Data Analysis

Before training the mode, a statistical study was carried out on the label distribution in the artificial dataset to understand its nature and to predict elements which might affect the detection performance in the future. In the total of 10,000 annotated frames, there is an evident imbalance among classes, as cars represent the biggest part of the annotated objects (11, 008 cars), while pedestrians (2,424 instances) and cyclists (1,652 instances) are for less presented (in a 5:1 ratio between cars and pedestrians). Besides, 32.6% of the frames had no annotated objects at all.

The analysis of the bounding box sizes found that pedestrians and cyclist objects are usually very small in size: 86.4% of pedestrian detections and 67.3% of cyclist detections have sizes less than 32x32 pixels, which has been demonstrated to be significantly harder to localize and detect using object detection models, such as YOLO. This result is methodological importance since, not taking it into consideration, one can erroneously assume that the low pedestrian and cyclist mAP is caused by problems with synthetic data generation instead of being a part of the object detection problem connected to the size of objects. The results per class are reported in this research.

3.6 Data Poisoning Detection

As the synthetic dataset had been passed through an external pipeline component, the label files were first validated for security before any training could take place. This was due to the risk that the data could be poisoned an instance where corrupting annotation files can corrupt a trained model without making any changes to the images themselves. Two validation techniques were used on every label file within the entire dataset (15, 084 annotated objects in five towns):

- Structural validation — checking each annotation line for the correct field count, a valid class identifier, normalised coordinate bounds (0 to 1), and a non-degenerate bounding box area.
- Statistical validation — comparing each town's per class label distribution against the distribution observed across the other towns, flagging any town whose class proportions deviated by more than four standard deviations, a pattern that would be expected if one town's labels had been selectively tampered with.

In order to ensure the proper functionality of the detection mechanism without providing just a trivial result such as “there are no anomalies,” the efficiency of the detector was tested by intentionally injecting two kinds of anomalies into a separate copy of a label file (which is not a part of the working data set): out of range coordinates (x_center = 1.5) and a wrong class ID (class ID = 7).

3.7 Dataset Preparation for YOLO Training

Before training, the decrypted synthetic dataset (10,000 image label pair samples from five simulation worlds) was arranged in the structure of the directory required by the YOLOv8 training module, where the images and labels were split into three sets: train set (70%, n = 7,000), validation set (15%, n = 1,500) and test set (15%, n = 1,500). Since there is no unique identification for frames among the five towns, files had to be named using a prefix representing the name of the town before copying them into the new directory structure, avoiding naming conflict that would have led to discarding a considerable part of the dataset. A seed of 42 was used in the randomization process to enable the reproducibility of the split as per the design of the project.

3.8 Model Training (Synthetic Baseline)

An object detection model (YOLOv8n, the smallest variant in the YOLOv8 family, comprising 3,006,233 parameters) was trained on the synthetic training split (7,000 images) for a maximum of 50 epochs, at an input resolution of 640×640 pixels, with a batch size of 8 and a fixed random seed (seed = 42) for reproducibility. Training was conducted on CPU (Intel Core i5-8250U) due to the absence of a compatible GPU on the available hardware; this increased training time substantially (approximately 50× slower than GPU execution) but did not affect the correctness of the resulting model, since all comparisons in this evaluation were conducted under identical hardware conditions. An early stopping criterion (patience = 15 epochs without validation improvement) was applied to avoid unnecessary computation once performance plateaued, and the best performing checkpoint (by validation mAP) was retained for subsequent evaluation.

3.9 mAP Evaluation Methodology

The performance of models was evaluated based on a specific test dataset that is different from the validation dataset used in the process of determining when to stop the training process in the order to avoid any bias due to validation leakage. There are two forms of mean average precision metrics evaluated, namely mAP50 (when there is an overlap of at least 50% between the ground truth bounding box and the predicted bounding box), and mAP50-95 (which is the average of the mAPs ranging from 50% to 95%). The more rigorous form is chosen as the main one to evaluate the model's performance, as it captures the localization ability better than mAP50.

3.10 Real Data Baseline: Selection and Preparation (KITTI)

Synthetic mAP alone provides little indication of practical utility without a real world point of comparison. The KITTI 2D Object Detection Benchmark was selected as an independent real world dataset for this purpose. KITTI was chosen deliberately as a dataset independent of the statistical sources used to calibrate the synthetic data generation pipeline (Stage 1, which draws on Waymo Open Dataset statistics), thereby avoiding circularity between the calibration source and the evaluation baseline and ensuring that any measured sim to real transfer reflects genuine generalization rather than statistical similarity introduced during privacy calibration.

The KITTI training split (7,481 image-label pairs) was downloaded and converted from KITTI's native annotation format to YOLO format. KITTI's absolute pixel coordinate bounding boxes (left, top, right and bottom) were converted to normalised Centre coordinate format, consistent with the label format of the synthetic dataset. Class mapping followed the same three class taxonomy used throughout the project: Car, Van and Truck were mapped to car; Pedestrian and Person_sitting were mapped to pedestrian; Cyclist was mapped to cyclist. The Tram, Misc and DontCare categories were excluded, consistent with the synthetic dataset's class structure. The converted dataset was split into train/validation/test sets using the same 70/15/15 ratio and fixed random seed (42) used for the synthetic dataset, yielding 5,236 training, 1,122 validation, and 1,123 test images.

A notable limitation of using KITTI as the real world baseline is the substantial difference in image resolution and aspect ratio between the two domains: CARLA generated synthetic images are 800×600 (4:3 aspect ratio), while KITTI images are approximately 1242×375 (a wide letterbox format). This is acknowledged explicitly as a confounder of the sim to real comparison: part of any observed performance gap between synthetic trained and real trained models may originate from this resolution and aspect ratio mismatch, rather than exclusively from the privacy preserving synthetic-data generation process itself.

3.11 Cross-Domain Evaluation Protocol

To quantify the utility retained by privacy preserving synthetic data in relation to real world data, an evaluation protocol that includes four experiments was developed. YOLOv8 models for object detection were trained separately the first using synthetic data generated in CARLA, while the second using real world data from KITTI. Both of the models were tested using the testing set of their respective domains and the testing set of the other domain. This results in four different train / test pairs, shown below.

#	Train Domain	Test Domain	Purpose
1	Synthetic	Synthetic	Can the model learn from this data at all?
2	Synthetic	Real (KITTI)	Sim to real transfer the primary utility metric
3	Real (KITTI)	Real (KITTI)	Upper bound with real data
4	Real (KITTI)	Synthetic	Reverse check on the domain gap

Reproducibility Information: Reproducibility information: model architecture – YOLOv8n; framework – Ultralytics 8.4.115; evaluation metrics – mAP50 (IoU ≥ 0.5), mAP50-95 (average IoU 0.5 – 0.95); test set size – synthetic 1,500 images, KITTI 1,123 images; random seed – 42; hardware – CPU (Intel i5-8250U).

3.12 FID Protocol

To complement the task level utility measurement, the Fréchet Inception Distance (FID) was computed to assess the image level statistical similarity between the synthetic (CARLA generated) and real world (KITTI) datasets. FID measures the distance between the feature distributions of two image sets, extracted using a pre-trained InceptionV3 network (2048 dimensional feature space), with lower values indicating greater statistical similarity. The feature extractor used was InceptionV3 (via the torchmetrics FrechetInceptionDistance implementation); images were resized to 299×299 and normalised prior to feature extraction. A sample size of 1,000 images per set (real and synthetic) was used, meeting the minimum threshold

recommended for reliable FID estimation; the real-image source was the KITTI test split and the synthetic-image source was the synthetic test split.

3.13 Evaluation Metrics Summary

In summary, this evaluation combines four complementary forms of measurement: **(i)** a security audit (data poisoning detection, including an empirical detector validation test); **(ii)** task level detection performance (mAP50 and mAP50-95, overall and per class); **(iii)** cross domain generalization (the four experiment protocol in Section 3.12); and **(iv)** distributional realism (FID). Together, these measurements constitute a complete utility evaluation that directly addresses the research aim stated.

4. Product and Solution Design

4.1 Introduction

This chapter is a description of the pipeline that was created as a tool to evaluate the utility of this particular component. During the implementation process, the method described in Chapter 3 was implemented as an actual set of Python scripts that can decrypt and validate the dataset, explore it, perform an audit, train and evaluate object detection models and measure utility.

4.2 Requirements

The requirements below were derived from the research aim in Section 1.5 and from the constraints of receiving data from an external pipeline component.

Functional requirements

- **FR1.** Decrypt every encrypted file (16-byte IV followed by AES-256-CBC ciphertext with PKCS7 padding) across all five towns.
- **FR2.** Verify the SHA-256 hash of every decrypted file against the town manifest and report any mismatch.
- **FR3.** Handle empty label files (background-only frames) as valid data without failure.
- **FR4.** Compute per-class counts, per-frame counts, the proportion of empty frames and the bounding-box size distribution.
- **FR5.** Validate every label file structurally and statistically, report a PASS/FAIL verdict, and demonstrably detect deliberately injected defects.
- **FR6.** Convert the synthetic dataset and the KITTI dataset to the same YOLO structure using the same 70/15/15 split and seed (42), with town-prefixed file names to avoid collisions.
- **FR7.** Train a synthetic-domain model and a real-domain model with identical hyperparameters.
- **FR8.** Evaluate the four train/test combinations on held-out test splits, reporting overall and per-class mAP50 and mAP50-95.
- **FR9.** Compute FID between 1,000 real and 1,000 synthetic images.

Non – functional requirements

- **Reproducibility.** Fixed seeds, recorded software versions and documented hyper parameters, so that all results can be regenerated.

- **Security.** The decryption key must not appear in source code or version control, and encrypted originals must be retained.
- **Integrity.** Decryption must be lossless and verifiable, and the label data must be audited before it is used for training.
- **Robustness.** Scripts must not crash on empty labels or on differences in annotation format.
- **Comparability.** Both domains must be processed with identical splits, hyper parameters and hardware.
- **Independence.** The real-world baseline must be independent of the data used to calibrate the generator.
- **Feasibility.** All steps must run on the available CPU-only hardware.
- **Transparency.** Per-class results and known confounders must be reported rather than hidden behind aggregate figures.

4.3 Component architecture within the complete solution

This component occupies the third stage of the group's four stage pipeline (Section 1.1). It consumes the encrypted synthetic dataset produced by the dataset generation component (Stage 2) and produces, as its output, a set of quantitative utility findings (mAP, FID, security audit results) that feed directly into the group's combined privacy utility argument, alongside the model level privacy cost findings produced by the privacy calibration component (Stage 1).

4.4 Design alternatives and decisions

Table 2 summarizes the main design alternatives that were considered and the decisions taken for this component.

Decision area	Alternative considered	Decision and rationale
Real-world baseline	Reuse the Waymo Open Dataset used to calibrate the generator	KITTI was chosen. An independent dataset avoids circularity between the calibration source and the evaluation baseline.
Experiment design	Only synthetic-to-real transfer	Four train/test combinations were used. They add a learnability check, a real-data upper bound and a reverse check of the domain gap.
Reporting level	Aggregate mAP only	Per-class and per-size reporting was adopted. Aggregate mAP would mask the weak pedestrian and cyclist results. mAP50-95 is the primary metric.
Evaluation split	Report on the validation split	A separate held-out test split is used, so early-stopping decisions cannot leak into the reported results.
Poisoning detection	Structural checks only	Structural and statistical (four standard deviations between towns) checks were combined, and the detector was proven with deliberately injected defects.
Detector model	Larger YOLOv8 variants	YOLOv8n (3,006,233 parameters) was used because training had to run on a CPU. Both domains use identical hyperparameters.
Realism metric	mAP only	FID was added as a complementary, task-independent measure, using 1,000 images per set.

Decision area	Alternative considered	Decision and rationale
Privacy budgets	Evaluate several epsilon values	Not possible with the available data, because epsilon is applied upstream of scene generation. A single configuration (epsilon = 1.8) is evaluated and the protocol is reusable.
Empty-label frames	Exclude them from the dataset	They are retained as valid background-only samples (32.6% of frames), and the scripts handle them explicitly.
File naming	Keep original frame identifiers	Town-prefixed names were used. Frame indices repeat across towns, so original names would silently overwrite files.
Key management	Hard-code the key in the scripts	The key is read at runtime from an environment-managed configuration file that is excluded from version control.

Table 2: Design Alternatives and Decision

4.5 Decryption Module

The decryption module (decrypt.py) implements AES 256 CBC decryption with PKCS7 padding removal, reading the 32 byte key from an environment variable rather than hardcoding it in source code. For every encrypted file across the five town directories, the module decrypts the file, computes its SHA 256 hash, and compares the hash against the town's reference manifest, reporting any mismatch immediately. Non encrypted label files are copied without modification. The module was designed so that encrypted originals are never deleted, allowing the decryption process to be safely repeated without re requesting the dataset.

4.6 Data Exploration Module

The exploration module iterates over every label file in the decrypted dataset, computing per class object counts, per frame object counts, the proportion of empty (background only) frames, and the pixel area distribution of each class relative to the native 800×600 image resolution.

These statistics informed the class imbalance and small object findings reported in Section 3.6, and directly justify per class and per size reporting adopted throughout this evaluation.

4.7 Poisoning Detection Module

The poisoning detection module (poison_detection.py) performs the structural and statistical validation described in Section 3.7 across every label file in the dataset, producing a per town class distribution summary, a list of flagged anomalies (if any), and an overall PASS/FAIL verdict. A companion script (prove_poison_detection.py) deliberately introduces two known defects into an isolated copy of a label file and re runs the detector against it, providing empirical evidence that the detector actively identifies tampering rather than passing all input unconditionally.

4.8 Dataset Conversion Module

Two conversion scripts were implemented. The first converts the decrypted synthetic dataset into the directory structure and data.yaml configuration required by the Ultralytics YOLOv8 framework, applying the town prefixed renaming and fixed seed 70/15/15 split described in Section 3.8. The second converts the downloaded KITTI dataset whose native annotation format uses absolute pixel coordinates in a different field order into the same YOLO format structure, applying the class mapping described in Section 3.11 and the identical 70/15/15 split and seed, ensuring structural consistency between the synthetic and real world datasets for the downstream cross domain evaluation.

4.9 Model Training Module

Two independent training runs were executed using the Ultralytics YOLO training API: one on the synthetic dataset (producing the synthetic_baseline checkpoint) and one on the converted KITTI dataset (producing the kitti_baseline checkpoint), using identical hyper parameters (50 epochs, image size 640, batch size 8, seed 42, early-stopping patience 15) to isolate the effect of the training data domain rather than training-configuration differences.

4.10 Cross Domain Evaluation Module

The cross evaluation module (cross_evaluation.py) loads both trained model checkpoints and iterates over the four train/test combinations described in Section 3.12, invoking the Ultralytics validation API against the relevant dataset configuration in each case and recording overall and per class mAP50 and mAP50-95 for each experiment, consolidated into the summary table presented in Chapter 5.

4.11 FID Computation Module

The FID module (compute_fid.py) loads 1,000 images from each of the real (KITTI) and synthetic test splits, resizes and normalises them, and streams them in batches through the torchmetrics Fréchet Inception Distance implementation (InceptionV3 backbone, 2048 dimensional feature space), reporting the final computed FID score alongside the number of images used per set, in line with the reporting requirement for small sample FID reliability.

4.12 Standards and professional practice

The following standards and professional practices were followed.

- **Cryptography.** AES – 256 as per NIST FIPS 197 [8] was applied for decryption key was removed from source code and version control.
- **Data Protection.** Motivation and design are aligned with GDPR [11] and the Personal Data Protection Act of Sri Lanka [12]. Evaluation uses synthetic imagery, and only real world data that is being used is the publicly available KITTI benchmark with license terms for academic research purposes.
- **Privacy.** The theoretical background of differential privacy is based on the work of Dwork and Roth [1]. This component uses the privatized output of stage 1 and does not degrade its quality.
- **Security practices.** Data provided by another component is considered untrusted and inspected for poisoning [9]. Poisoning detector is tested using a test with known defect and encrypted original images are stored to make further processing repeatable.
- **Academic integrity.** Generative AI use is disclosed in Appendix B in accordance with the module policy.

5. Implementation Integration and Testing

5.1 Development environment and technology stack

All methods related to data processing, security checks and model training, as presented in this study, have been done in Python (3.10 / 3.11) in a separate conda environment. The object detection model training and prediction tasks have been executed using the Ultralytics YOLOv8 framework, based on PyTorch deep learning library. Decryption of the cipher text has been done through the usage of PyCryptodome library whereas statistical computations and data exploration have been done with the help of Numpy, Scipy and Pandas.

Because of hardware limitations particularly, lack of a suitable GPU driver for the development machine, the model was trained using CPU (Intel Core i5-8250U) instead of GPU. Although this did not impact the accuracy of the output, it significantly increased the training time compared to training with GPU acceleration (about 50 times slower).

5.2 Implementation Challenges

Several practical challenges were encountered and resolved during implementation:

- As a consequence of the Ultralytics default project folder generation, the trained model checkpoint files needed to be explicitly checked within the file system in order to make sure that the right paths were used in the cross evaluation script.
- While preparing the datasets, care had to be taken to take into account those image labels which were empty in order not to crash the script, because 32.6% of synthetic frames and 475 out of 1,500 synthetic test images contained no objects at all.
- Converting KITTI's native annotation format (absolute pixel coordinates in a non-YOLO field order) to the normalised YOLO format required careful coordinate transformation logic and class mapping, with validation against the dataset's documented format.
- The downloading, registration and extraction of the KITTI benchmark dataset (approximately 12 GB of images) within the set limits of storage space and time necessitated the proper handling of interim raw data files.
- Machine learning training of all models used took place using CPU only as there was no GPU available, which implied overnight training sessions and taking special care of the computer’s sleep mode settings.

5.3 Integration with the other components

This element is both the consumer of the data generation stage and the producer of the information needed by other stages.

Connection to stage 2 (Dataset Generation). The data generation element produces encrypted by AES – 256 images (RGB, depth, semantic segmentation and class mask at 800 x 600 resolutions), plain text YOLO label files and SHA – 256 manifests for each of the five towns. These are consumed by the decryption element, and the manifest comparison showed that all 40,000 images arrived without issues. Of these, only the RGB images and labels are needed for the detector training. As frame IDs overlap between towns, town based renames were made at the receiving end in order to incorporate five towns into one dataset.

Connection to stage 1 (Privacy Calibration). The synthetic data was created based on privatized with Laplace mechanism parameters at epsilon = 1.8; that is the reason why all the results depend on this particular parameterization. This way, the model level cost found in stage 1 (DP – SGD: 0.17 AUC loss, with AUC loss of membership inference attack from 0.65 to 0.50) connects with the data level cost found here table 5 (Connecting Findings to the Group Pipeline).

Interface with stage 4 (Risk and Compliance). The verdict of poisoning audit (PASS, no anomalies found out of 15,084 boxes), the encryption and key management controls and the

results of integrity verification can be used as proof of application of controls for the risk and compliance module at the handover of Stages 2 and 3.

Integration Testing. The integration was checked by means of decryption test (40,000 out of 40,000 files, zero errors), conversion test (7,000 / 1,500 / 1,500 split synthetics) and successful completion of both training and all four evaluation runs. If more datasets are generated by the generation stage for different epsilons, they could pass the same modules untouched.

6. Testing and Evaluation

6.1 Testing Procedure

This chapter reports the results of applying the Chapter 3 methodology: security audit, synthetic only evaluations, real data baseline, cross domain evaluation and FID computation.

6.2 Poisoning Detection Results

The structural and statistical checks on the full dataset (15,084 object instances, five towns) found zero anomalies and no significant between town class distribution deviation, giving a PASS verdict. A deliberately poisoned label file (x_center = 1.5, class ID 7) was correctly flagged (3 anomalies), confirming that the detector actively identifies tampering and that the dataset was not corrupted before training.

6.3 Synthetic Data Model Performance

On the held out test subset (1,500 images, 2,367 instances), excluded from training and validation, the synthetic baseline achieved mAP50 = 0.3655 and mAP50-95 = 0.2273. Per class AP50-95 was 0.3981 (Car), 0.1882 (Cyclist) and 0.0955 (Pedestrian); Car performance was about 4.2 times that of Pedestrian, reflecting the roughly 5:1 class imbalance and small object sizes found in the exploratory analysis.

6.4 Real Data Baseline (KITTI) Preparation Results

KITTI conversion produced 7,481 image label pairs (5,236 train / 1,122 validation / 1,123 test; 70/15/15, seed = 42) in the same layout as the synthetic dataset. A second YOLOv8n model (kitti_baseline) was trained with identical hyper parameters, isolating the effect of the training data domain.

6.5 Cross Domain Evaluation Results

Table 3 summarizes the four experiments cross domain protocol described in Section 3.12.

Experiment	mAP50	mAP50-95	Car AP50	Pedestrian AP50	Cyclist AP50
Synthetic → Synthetic	0.3655	0.2273	0.6076	0.1793	0.3096
Synthetic → Real (KITTI)	0.0889	0.0343	0.2614	0.0051	0.0001
Real (KITTI) → Real (KITTI)	0.8500	0.5531	0.9519	0.7769	0.8213
Real (KITTI) → Synthetic	0.0507	0.0199	0.1273	0.0079	0.0171

Table 3: Full Cross-Domain Evaluation Results

Synthetic-to-real mAP50 (0.0889) represents 10.46% of the real-to-real mAP50 (0.8500), that is, (0.0889 ÷ 0.8500) x 100, the quantifiable price of privacy on the data side (Section 1.3). However, the cost distribution is not even: while Car kept about 27.5% of real data AP50 (0.2614 vs. 0.9519), Pedestrian (0.0051) and Cyclist (0.0001) dropped, reflecting class imbalance and small objects (Section 3.6). Conversely, the reverse transfer (Real → Synthetic, mAP50 = 0.0507) indicates that the gap between domains goes both ways (Section 3.11), and the consistent ordering Real → Real > Synthetic → Synthetic > cross domain pairs underpins protocol internal validity.

6.6 FID Results

The computed FID between the synthetic and real (KITTI) test image sets was 115.99 (n = 1,000 images per set).

Range	Interpretation
< 50	Reasonable similarity
50 – 200	Moderate difference
> 200	Very different

Table 4: FID Interpretation against Reference Scale

The FID between the synthetic and real (KITTI) test sets was 115.99 (n = 1,000 per set), within the "moderate difference" range (< 50 reasonable similarity, 50–200 moderate, > 200 very different), yet this did not translate into proportional utility (10.46% mAP retention, near zero Pedestrian and Cyclist AP). FID alone therefore does not predict object level detection performance; mAP and FID should be used together.

6.7 Consolidated Testing Summary

Test Item	Expected Result	Actual Result	Status
Dataset decryption	All files decrypted	40,000/40,000, 0 failed	Passed
Poisoning audit and deliberate test	Clean passes; defects flagged	PASS; 3 anomalies flagged	Passed
YOLO and KITTI conversion	70/15/15 split	7,000/1,500/1,500; 7,481 converted	Passed
Training and cross-domain evaluation	Runs complete; logical mAP order	All completed; order as expected	Passed
FID computation	≥ 1,000 images per set	FID = 115.99	Passed

Table 5: Functional and Experimental Test Results

6.8 Deployment and Operation

The solution itself is implemented as a chain of batch style evaluation programs and is not running as a service. The chain should be executed in the following manner:

1. The decryption key should be set in the configuration file managed by the environment (and never committed to version control), and the decrypted.py program should be executed to restore the images and check SHA 256 hashes.
2. The exploration program to obtain class, empty frame and object size statistics should be run.
3. The poison detection program (poison_detection.py) and to make sure it works the companion prove_poison_detection.py program should be run.
4. The two scripts for the creation of the synthetic and KITTI dataset in YOLO format in 70/15/15 distribution with seed 42 should be executed.
5. The synthetic baseline and kitti_baseline should be trained using exactly the same parameters (50 epoches, images size 640, batch size 8 and patience 15).
6. The cross domain evaluation (cross_evaluation.py) and FID computation (compute_fid.py).

6.9 Discussion of Results

The synthetic dataset trains a functional detector (mAP50=0.3655) and all tests passed, but utility transfers poorly to real data (10.45% retention), mainly for Pedestrian and Cyclist. Two confounders apply: the CARLA (800x600) vs. KITTI (1242 x 375) resolution and aspect ratio mismatch, so part of the drop is not due to privacy preserving generation itself and cannot be fully isolated; and CPU only training, which increased run time but not validity since all comparisons used identical hardware.

7. Discussion

7.1 Interpretation of findings

The synthetic dataset trains a functional detector (mAP50=0.3655) and all tests passed, but utility transfers poorly to real data (10.45% retention), mainly for Pedestrian and Cyclist. Two confounders apply: the CARLA (800x600) vs. KITTI (1242 x 375) resolution and aspect ratio mismatch, so part of the drop is not due to privacy preserving generation itself and cannot be fully isolated; and CPU only training, which increased run time but not validity since all comparisons used identical hardware.

7.2 Research contribution

This part (i) measures the cost of privacy (10.46% retention) in terms of data; (ii) establishes a relationship between the pedestrian and cyclist collapse and class imbalance and small objects; (iii) proves that there is a discrepancy between FID and mAP; (iv) employs an independent methodology, in which the KITTI is separate from the Waymo calibration source and there are four experiments, among which are an upper bound on the real data and a reverse transfer; (v) introduces a protocol applicable to other epsilon values; and (vi) provides a label integrity audit.

7.3 Product contribution

A reproducible set of Python modules (decryption and integrity, exploration, poisoning detection, conversion, training, cross-domain evaluation and FID; Section 4.4) that considers the issue of empty labels, avoids naming conflicts, hides secrets from version control and allows repeating on any future dataset of the same format.

7.4 Limitations and threats to validity

There is only one privacy parameter (epsilon = 1.8), so the privacy–utility curve cannot be constructed; the KITTI–CARLA resolution and aspect-ratio difference is an uncontrolled confounder; all the experiments were run on CPU.

7.5 Implications

There is a quantifiable cost to privacy at both data and model levels (Table 5). So statements must include both aspects. Generators should be evaluated based on their capacity to faithfully represent the vulnerable road users and this dataset by itself is not fit for deployment. The FID is not enough to prove the utility of the dataset; a task-level evaluation on actual data is required per class. 10.46% includes the gap between the rendering and resolution discrepancy and the privatisation effect, thus it is an upper bound on the cost of privacy when epsilon = 1.8.

8. Conclusion

8.1 Conclusion

This work aimed at empirically evaluating whether the privacy preserving synthetic driving dataset, created using a CARLA simulator which was calibrated by differential privacy parameters, is still useful enough to train an object detection model for autonomous vehicles as compared to the real world dataset. It has been carried out using an entire utility evaluation pipeline consisting of decryption and validation of data, data exploration and detection of poisoned data and its empirical validation, training the YOLOv8 model, conducting four experiments on cross domain evaluation and calculating Fréchet Inception Distance. This module has helped in completing the privacy utility trade off of the research work carried out by the group.

The results show that the synthetic dataset is capable of training a functional perception model (mAP50 = 0.3655 on synthetic data), but retains only 10.46% of real data detection performance when transferred to real world imagery, with this utility cost concentrated disproportionately on the pedestrian and cyclist classes due to class imbalance and small object prevalence in the dataset. The data poisoning audit confirmed, with empirical validation, that the received dataset was free of tampering, and the FID score of 115.99 indicated moderate, rather than extreme, distributional dissimilarity between synthetic and real imagery a finding that, combined with the mAP results, demonstrates that aggregate visual realism alone does not guarantee object detection utility.

This component plays an important role in the broader group project, as it provides the data level cost measurement that completes the project's central privacy utility argument, pairing directly with the model level cost (DP – SGD) quantified by the privacy calibration component.

8.2 Limitations and Future Improvement

Even though all objectives have been met, there are certain limitations associated with this evaluation. Specifically, the synthetic dataset has been created using only one value for the privacy parameter (epsilon=1.8). This is because of the fact that in the existing pipeline design, the privatization of configuration parameters occurs upstream from the scene generation and not a dataset itself. Therefore, it was impossible to construct a privacy utility curve as part of the current research. The real world dataset (KITTI) differs significantly from the synthetic dataset in terms of resolution and aspect ratio. This creates another confounding variable that could not be controlled. All experiments have been carried out using the CPU as the GPU was not available.

In terms of future improvements for this research, in case of running the dataset creation process for multiple calibration epsilon values, the methodology introduced in this report could be simply replicated for each generated dataset in order to build up a privacy versus utility trade off curve.

An alternative real world baseline with similar aspect ratio like the Cityscapes dataset would help in reducing the problem with resolution mentioned in section 3.11. Domain adaptation methods, including fine tuning of the synthetically trained model on a few real world samples, could be also tested as an appropriate way of closing the simulation real gap, together with GPU acceleration training.

8.3 Final Remarks

This component's findings integrate with the group's broader investigation into the privacy utility trade off of the proposed pipeline, as summarized in Table 6.

Level	**Component**	**Finding**
Data	Utility Evaluation (this work)	Synthetic data retains 10.46% of real data detection utility.
Model	Privacy Calibration (DP-SGD)	DP-SGD costs 0.17 AUC and defeats membership inference (attack AUC 0.65 → 0.50).

Table 6: Connecting Findings to the Group Pipeline

Both these points together prove that there is an observable price to be paid in terms of utility in the processes of data creation and model training by the application of privacy preserving mechanisms, thus proving the point made in the project that privacy preserving techniques have a cost associated with them and this element has measured it.

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